

What are Date Functions?

Date functions in Power BI (DAX) are used to:

- Work with **dates and time**
- Create **calendar tables**
- Perform **time-based analysis** (daily, monthly, yearly)

1. Create a Date (Basic)

```
DateColumn = DATE(2026, 4, 23)
```

Creates a date → **23-Apr-2026**

2. Extract Year, Month, Day

```
Year = YEAR(OldTable[Date])  
Month = MONTH(OldTable[Date])  
Day = DAY(OldTable[Date])
```

Extracts parts of a date

3. Current Date & Time

```
TodayDate = TODAY()  
CurrentDateTime = NOW()
```

TODAY() → current date

NOW() → date + time

4. Add / Subtract Days

```
NextDate = OldTable[Date] + 7  
PreviousDate = OldTable[Date] - 7
```

Adds or subtracts days

5. Start and End of Month

```
StartOfMonth = STARTOFMONTH(OldTable[Date])  
EndOfMonth = ENDOFMONTH(OldTable[Date])
```

Useful for monthly reports

6. Date Difference

```
DateDiff = DATEDIFF(OldTable[StartDate], OldTable[EndDate], DAY)
```

Units can be:

- DAY
- MONTH
- YEAR

7. Create Calendar Table

```
CalendarTable = CALENDAR (DATE (2020,1,1), DATE (2026,12,31))
```

Creates a full date table

8. Auto Calendar Table

```
CalendarAuto = CALENDARAUTO()
```

Automatically creates date range from your data

9. Format Date

```
FormattedDate = FORMAT (OldTable[Date], "DD-MM-YYYY")
```

Converts date into text format

10. Month Name & Day Name

```
MonthName = FORMAT (OldTable[Date], "MMMM")
DayName = FORMAT (OldTable[Date], "DDDD")
```

Example:

- January
- Monday

Date Table is Mandatory

For proper reporting (Year, Month, Trends), always create a **Calendar Table**

Mark as Date Table

After creating:

Go to **Model View** → **Mark as Date Table**

Relationship

Connect:

```
Calendar[Date] → Sales[Date]
```

Common Use Cases

Task	Function
Get current date	TODAY()
Extract year	YEAR()
Find difference	DATEDIFF()
Create calendar	CALENDAR()
Format date	FORMAT()

Practice Example

```
CalendarTable =
ADDCOLUMNS (
    CALENDAR (DATE (2023,1,1), DATE (2026,12,31)),
    "Year", YEAR ([Date]),
    "Month", FORMAT ([Date], "MMMM"),
    "MonthNo", MONTH ([Date])
)
```

What is a New Table?

A **New Table** in Power BI is created using **DAX (Data Analysis Expressions)** from existing data.

It is used for:

- Filtering data
- Selecting specific columns
- Creating summary tables
- Removing duplicates

Steps to Create a New Table

1. Go to **Modeling Tab**
2. Click on **New Table**
3. Enter a DAX formula in the formula bar

1. Copy an Existing Table

Create an exact duplicate:

```
NewTable = OldTable
```

Copies all rows and columns

2. Create Table with Selected Columns

Use only required columns:

```
NewTable =
SELECTCOLUMNS (
```

```
OldTable,  
"Department", OldTable[Department],  
"Salary", OldTable[Salary]  
)
```

Creates a table with only chosen columns

3. Create Table with Filter Condition

Filter rows based on a condition:

```
NewTable =  
FILTER(  
    OldTable,  
    OldTable[Salary] > 50000  
)
```

Includes only rows where salary > 50,000

4. Create Table with Unique Values

Remove duplicates:

```
DeptTable =  
DISTINCT(OldTable[Department])
```

Returns unique department names

5. Create Summary Table (Group By)

Aggregate data:

```
SummaryTable =  
SUMMARIZE(  
    OldTable,  
    OldTable[Department],  
    "TotalSalary", SUM(OldTable[Salary])  
)
```

Shows total salary per department

Important Points

- New tables are **static** (created during data refresh)
- They do **not change dynamically** like measures
- Used mainly for **data modeling and structure**

When to Use What

Requirement	Function
Copy full table	OldTable
Select columns	SELECTCOLUMNS
Apply filter	FILTER
Remove duplicates	DISTINCT
Group & summarize	SUMMARIZE

CALCULATE Function (Most Important)

Purpose:

Changes the **filter context** and evaluates an expression.

Syntax:

```
CALCULATE(<expression>, <filter1>, <filter2>, ...)
```

Example:

```
Total Sales = SUM(Sales[Amount])

Sales in 2024 =
CALCULATE(
    [Total Sales],
    Sales[Year] = 2024
)
```

CALCULATE is the **base function** used with all time intelligence formulas.

DATESBETWEEN

Purpose:

Returns data between **two specific dates**

Syntax:

```
DATESBETWEEN(<dates>, <start_date>, <end_date>)
```

Example:

```
Sales Between Dates =
CALCULATE(
    [Total Sales],
    DATESBETWEEN(
        Date[Date],
        DATE(2024,1,1),
        DATE(2024,3,31)
    )
)
```

Used when you want **custom date range filtering**

MTD (Month-To-Date)

Purpose:

Calculates value from **start of month → current date**

Formula:

```
MTD Sales =  
CALCULATE(  
    [Total Sales],  
    DATESMTD(Date[Date])  
)
```

Example:

If today is **15 March**, it calculates:

1 March → 15 March

QTD (Quarter-To-Date)

Purpose:

Calculates value from **start of quarter → current date**

Formula:

```
QTD Sales =  
CALCULATE(  
    [Total Sales],  
    DATESQTD(Date[Date])  
)
```

Example:

If current quarter = Jan–Mar

It calculates: 1 Jan → today

YTD (Year-To-Date)

Purpose:

Calculates value from **start of year → current date**

Formula:

```
YTD Sales =  
CALCULATE(  
    [Total Sales],
```

```
DATESYTD(Date[Date])
)
```

Example:

If today is 20 Aug 2024

It calculates: 1 Jan → 20 Aug

Important Requirement

All time intelligence functions need a **proper Date Table**

Create Date Table:

```
Date Table = CALENDAR (DATE (2020,1,1) , DATE (2025,12,31))
```

Then:

- Mark as **Date Table**
- Create relationship with your **Sales table**

Quick Comparison

Function	Meaning	Range
MTD	Month to Date	Start of Month → Today
QTD	Quarter to Date	Start of Quarter → Today
YTD	Year to Date	Start of Year → Today
DATESBETWEEN	Custom Range	Any dates